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JSContact Profile for RESTful Provisioning Protocol (RPP)

Abstract

This document defines a JSContact usage profile, conforming to the rules described in [I-D.ietf-calext-jscontact-profiles], for the RESTful Provisioning Protocol (RPP). The JSContact Profile for RPP specifies how the JSContact format defined in [RFC9982] can be used as a standardized representation of contact information within RPP operations, enabling interoperability and consistency across RPP implementations that manage contact data.

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1. Introduction

JSContact [RFC9982] is a flexible, JSON-based format for representing contact information. Its flexibility allows the same data to be expressed in multiple valid ways, which can hinder interoperability in a provisioning context where data must be stored and interpreted consistently.

This document defines a JSContact usage profile for the RESTful Provisioning Protocol (RPP), restricting the set of JSContact constructs that RPP clients and servers are required to support when provisioning contact information.

1.1. Relationship to the RDAP JSContact Profile

The Registration Data Access Protocol (RDAP) [RFC9083] provides read-only access to registration data managed by RPP. Both protocols use JSContact for contact representation, but with profiles suited to their respective roles.

The RDAP JSContact profile [I-D.ietf-regext-rdap-jscontact] defines the constructs used when serving registration data to read-only consumers. The RPP profile, defined in this document, is a subset of these constructs. All JSContact constructs permitted by the RPP profile can be used for RDAP.

The converse does not hold. RDAP supports additional JSContact constructs that are useful in the read-only context of RDAP responses but that may hinder interoperability. These constructs are not present in RPP requests and responses.

2. Terminology

In this document, the following terminology is used.

RESTful Provisioning Protocol or RPP - The protocol described in this document.

URL - A Uniform Resource Locator as defined in [RFC3986].

RPP client - An HTTP user agent performing an RPP request.

RPP server - An HTTP server responsible for processing requests and returning results in any supported media type.

3. Conventions Used in This Document

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT","SHOULD", "SHOULD NOT", "RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in [RFC2119].

4. Using JSContact in RPP

4.1. JSContact Profile for RPP

Since JSContact is a generalized representation of contact data, many of its capabilities are not optimized for interoperable use in RPP payloads. This document therefore defines a usage profile for JSContact that simplifies implementation for both RPP clients and RPP servers.

The JSContact profile for RPP is compliant with the rules described in [I-D.ietf-calext-jscontact-profiles]. The profile properties are formally listed in Section 4.1.12. All types mentioned in this section refer to the JSContact specification [RFC9982].

4.1.1. Version

The Card "version" member value MUST be "2.0".

4.1.2. Kind

The Card "kind" member value MUST be "individual" (default) or "org" to represent an individual or an organization, respectively.

4.1.3. Language

The Card "language" member SHOULD be set when localizations are specified (i.e., when the "localizations" member is present and non-null).

4.1.4. Name

The Card "name" member MUST include only the "full" member and MUST NOT include the "components" member.

4.1.5. Organizations

The Organization type MUST include only the "org" member.

4.1.6. Addresses

The Address type MUST only use the "components" member.

The AddressComponent MUST include only the "kind" and "value" members. The "kind" member value MUST only use "name", "locality", "region", or "postcode".

When both an internationalized and a localized version of an address are used, the localized version MUST be placed in the `localizations` map keyed by an appropriate [BCP47] language tag.

4.1.7. Emails

The EmailAddress type MUST include only the "address" member.

4.1.8. Phones

The Phone type MUST include the "number" member and MAY include the "features" member. When the "features" member is present, its values MUST be "voice" or "fax". When the "features" member is absent, the phone number is assumed to be a voice number.

4.1.9. Links

The Link type MUST include the "uri" member and MAY include the "kind" member. When the "kind" member is present, its value MUST be "contact".

4.1.10. Map Keys

A JSContact map key MUST comply with the Id type definition in [RFC9553]. RPP implementations MUST use the following predefined keys:

- "org" in the "organizations" map for a single preferred organization.
- "addr" in the "addresses" map for a single preferred postal address.
- "email" in the "emails" map for the preferred email address.
- "voice" in the "phones" map for the preferred voice number.
- "fax" in the "phones" map for the preferred fax number.
- "url" in the "links" map for a preferred contact URL; the "kind" member of the Link object MUST NOT be set.
- "contact-uri" in the "links" map for a contact URI; the "kind" member of the Link object MUST be set to "contact".

When internationalized and localized versions of an organization, postal address, or email exist, the internationalized version MUST be included under the predefined map key, while the localized version MUST be placed in the "localizations" map as described in Section 4.1.11.

4.1.11. Localizations

If present, localized variants of name, organization, postal address, and email MUST be added to the "localizations" map. RPP implementations MUST expand all localizations, meaning a nested PatchObject key of the form "{key1}/{key2}/.../{keyN}" MUST NOT be used.

The number of provided localizations MUST NOT exceed the number of languages supported by the RPP server. The language of each localization MUST be indicated by a [BCP47] language tag as the key in the "localizations" map.

The following example of a Card includes a localized version of the postal address:

```
{
  "@type": "Card",
  "version": "2.0",
  "language": "en",
  "name": {
    "full": "Joe User"
  },
  "organizations": {
    "org": { "name": "Example Org" }
  },
  "addresses": {
    "addr": {
      "components": [
        { "kind": "name",     "value": "Main Street 1" },
        { "kind": "locality", "value": "Ludwigshafen am Rhein" },
        { "kind": "region",   "value": "Rhineland-Palatinate" },
        { "kind": "postcode", "value": "67067" }
      ],
      "countryCode": "DE"
    }
  },
  "localizations": {
    "de": {
      "addresses": {
        "addr": {
          "components": [
            { "kind": "name",     "value": "Hauptstraße 1" },
            { "kind": "locality", "value": "Ludwigshafen am Rhein" },
            { "kind": "region",   "value": "Rheinland-Pfalz" },
            { "kind": "postcode", "value": "67067" }
          ],
          "countryCode": "DE"
        }
      }
    }
  }
}
```

Figure 1: Example of handling localizations in JSContact

4.1.12. Profile Properties

The properties of the JSContact profile for RPP registered in the JSContact Profile registry as described in Section 5.1 are:

| Property | Context | Restricted | Restricted Enum |

		Attributes	Values
language	Card		
kind	Card		individual, org
name	Card		
full	Name	mandatory	
organizations	Card		
name	Organization		
addresses	Card		
full	Address		
countryCode	Address		
components	Address		
kind	AddressComponent		name, locality, region, postcode
value	AddressComponent		
emails	Card		
address	EmailAddress		
phones	Card		
features	Phone		voice, fax
number	Phone		
links	Card		
kind	Link		contact
uri	Link		
localizations	Card		yes

Table 1

Client validation MUST return an error if any JSContact properties other than those listed above are present.

5. IANA Considerations

5.1. JSContact Profile Registry

IANA is requested to register the following entry in the "JSContact Profile" registry [I-D.ietf-calext-jscontact-profiles]:

Name: rpp Profile Version: 1.0 Reference: This document

TODO

6. Internationalization Considerations

TODO

7. Security Considerations

TODO

8. Change History

8.1. Version 00 to 01

- Fixed duplicate text (Issue #14)

9. Normative References

[BCP47] Best Current Practice 47, <https://www.rfc-editor.org/info/bcp47>.
At the time of writing, this BCP comprises the following:

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Phillips, A., Ed. and M. Davis, Ed., "Tags for Identifying Languages", BCP 47, RFC 5646, DOI 10.17487/RFC5646, September 2009, <https://www.rfc-editor.org/info/rfc5646>.

[I-D.ietf-calext-jscontact-profiles] Stepanek, R. and M. Loffredo, "Protocol-Specific Profiles for JSContact", Work in Progress, Internet-Draft, draft-ietf-calext-jscontact-profiles-15, 27 July 2026, <https://datatracker.ietf.org/doc/html/draft-ietf-calext-jscontact-profiles-15>.

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[RFC2119] Bradner, S., "Key words for use in RFCs to Indicate Requirement Levels", BCP 14, RFC 2119, DOI 10.17487/RFC2119, March 1997, <https://www.rfc-editor.org/info/rfc2119>.

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[RFC9553] Stepanek, R. and M. Loffredo, "JSContact: A JSON Representation of Contact Data", RFC 9553, DOI 10.17487/RFC9553, May 2024, <https://www.rfc-editor.org/info/rfc9553>.

[RFC9982] Stepanek, R., "JSContact Version 2.0: A JSON Representation of Contact Data", RFC 9982, DOI 10.17487/RFC9982, May 2026, <https://www.rfc-editor.org/info/rfc9982>.

Acknowledgements

TODO

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